Function & performance

Higher · Design · Worksheet

D&M WITH MR KELLY

Name _____ Class _____ Date _____

By the end of this worksheet I can…

- ☐ describe how function influences the design of a product
- ☐ explain primary and secondary functions with examples
- ☐ describe how performance and fitness for purpose influence design
- ☐ describe maintenance issues associated with products
- ☐ explain the influence of planned obsolescence on manufacturers, consumers and the environment
- ☐ explain value for money

I DO — Worked example

Read how a full-mark answer is built

A folding camping chair and a fixed garden recliner.

1 Describe how function and safety have influenced the design of these chairs. (5 marks)

Points that earn marks from the marking instructions: one ✓ per point

- ✓ Function: camping chair folds flat to fit in a car boot
- ✓ Function: cup holder in armrest keeps drinks within reach
- ✓ Function: recliner back adjusts to several positions
- ✓ Safety: wide leg spread keeps the chair stable on uneven ground
- ✓ Safety: locking catch stops the frame folding while in use

Why marks are not awarded

- ✗ Naming a function with no description
- ✗ Only one factor covered: max 4

WE DO — Guided practice

Use the starters and word bank to help you

2 Describe planned obsolescence and give an example of its influence on the design of products. (2 marks)

Sentence starters

Description: …

Example: …

Word bank

outdated · replaced · become · cannot · limited · designed · description · batteries

Tip: 2 marks = 2 separate, developed points.

YOU DO **On your own** No notes. Check your answers afterwards.

3 Quick check: tick one box for each.

a) A razor with replaceable heads compared with a disposable razor has...
☐ Shorter life expectancy ☐ Longer life expectancy ☐ No maintenance ☐ No secondary function

b) Making a phone battery hard to replace mainly affects which factor?
☐ Aesthetics ☐ Maintenance ☐ Target market ☐ Anthropometrics

c) A torch that also works as a bottle opener. The bottle opener is...
☐ Primary function ☐ Secondary function ☐ Performance ☐ Aesthetics

d) Which is a benefit of planned obsolescence for a manufacturer?
☐ Fewer sales ☐ Predictable repeat sales ☐ No need for R&D ☐ Better reputation guaranteed

4 Key terms: write the letter of the matching description next to each term.

1. **Primary function**	____	**A** Who maintains it, what tasks are needed, and how easily parts can be accessed or replaced.
2. **Secondary function**	____	**B** Designing products with a limited life so they will be replaced.
3. **Fitness for purpose**	____	**C** Additional jobs that add value to the product.
4. **Maintenance consideration**	____	**D** How well a product meets its intended use, user, environment and life expectancy.
5. **Planned obsolescence**	____	**E** The main job a product is designed to do.

5 Explain what is meant by ‘Style obsolescence’. (2 marks)

6 Explain what is meant by ‘Value for money’. (2 marks)

Check and reflect

Mark your answers with the answer sheet. Go back to the checklist on the site and re-rate each “I can” statement above. Which one still needs work?